

PROBLEMS FOR THE 16th INTERNATIONAL TOURNAMENT OF YOUNG MATHEMATICIANS

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Notation

$\mathbb{N} = \{1, 2, 3, \dots\}$	set of natural numbers (positive integers)
$\mathbb{Z}_n = \{0, 1, \dots, n-1\}$	ring of residue classes modulo n
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	sets of integer, rational, real and complex numbers
$\mathbb{R}_+$ and $\mathbb{R}_{\geq 0}$	sets of positive and non-negative real numbers
$\mathbb{R}^n$	n -dimensional real space
$\mathbb{F}_q$	finite field of size $q = p^k$
$\mathbb{P}(E)$	probability of an event E
$]a, b[$ and $[a, b]$	open and closed interval from a to b
$ x $	absolute value of x
$\#M$ or $ M $	cardinality of a set M
$M \times N$	Cartesian product of two sets or graphs M and N
$M \cup N, M \cap N, M \setminus N$	union, intersection and difference of two sets M and N

1. Right Sets

1. A set of points on the plane is called *right* if, for any two-coloring of it, there is at least one right triangle with vertices of the same color. Determine whether the following sets are right:
- circle;
 - boundary of an equilateral triangle;
 - boundary of a regular n -gon;
 - parabola;
 - $\mathbb{Z}^2$.

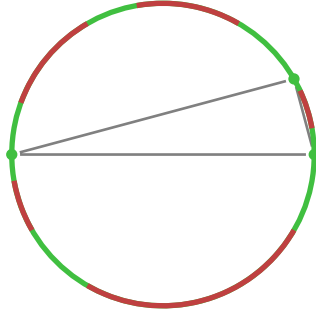


FIGURE 1. A two-coloring of a circle having a right triangle with vertices of the same color.

2. Describe the real functions $f(x)$ defined on a finite or infinite interval whose graphs are right sets.
3. Try to classify right sets.
4. Introduce also the concepts of *isosceles*, *equilateral*, *square* and *rectangular* sets and explore similar questions.
5. Formulate and investigate the problem in higher dimensions.

2. Nearest Points

Let S be one of the following shapes in the real plane:

- a) a unit square;
- b) a regular triangle with side length 1;
- c) a unit circle;
- d) an ellipse;
- e) a regular polygon with side length 1;
- f) a convex polygon of diameter (the largest distance between any pair of points of the polygon) d ;
- g) a non-convex polygon.

Let $n > 1$ and k be positive integers. Given n points $P_1, \dots, P_n$ inside S or on its boundary, consider the following functions:

$$f_k(P_1, \dots, P_n) = a_1^k + \dots + a_n^k \quad \text{and} \quad h_k(P_1, \dots, P_n) = b_1^k + \dots + b_n^k$$

where a_i and b_i are the distances from P_i to the nearest and farthest point, respectively, among $P_1, \dots, P_{i-1}, P_{i+1}, \dots, P_n$.

1. When S is a unit square, show that $f_2(P_1, \dots, P_n) \leq 4$, for any n and any $P_1, \dots, P_n$.
2. Find upper bounds for f_k and h_k in the cases a) – g).
3. The same question for other types of functions depending on a_i and b_i . For instance, you might choose another symmetric polynomial.
4. Consider other shapes S .
5. Formulate and investigate the problem in higher dimensions.

3. Cutting Polygons

Let x be a positive real number. Denote by $F(x)$ the least number of (not necessary convex) polygons into which one can cut a $1 \times x$ rectangle and form with obtained pieces a square of

area x (rotation is allowed but not turning over). If it is not possible to do so, then we put $F(x) = \infty$.

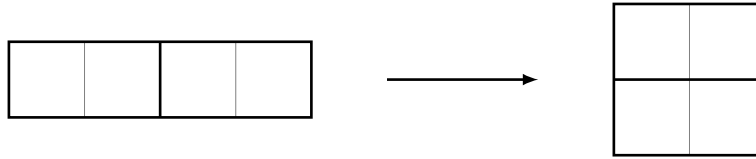


FIGURE 2. An example showing that $F(4) = 2$.

1. Prove that $F(9/4) = F(16/9) = 2$.
2. Prove that $\max\{F(2), F(3)\} \leq 3$.
3. Compute $F(2)$ and $F(3)$.
4. Denote $\mathcal{F} = \{x \in \mathbb{R} \mid F(x) \neq \infty\}$.
 - a) Prove that $\mathcal{F}$ is not bounded from above.
 - b) Find $\lim_{x \rightarrow \infty, x \in \mathcal{F}} F(x)$ or prove that the limit does not exist.
 - c) For a set A denote by A_+ the subset of positive numbers in A . Check whether the following sets are subsets of $\mathcal{F}$:
 - a) $[1, 4]$; b) $\mathbb{Z}_+$; c) $\mathbb{Q}_+$; d) $\mathbb{R}_+$.
5. Try to compute $F(x)$ for infinite families of positive real numbers x .
6. Give lower and/or upper bounds for $F(x)$.
7. Introduce and estimate a similar function, when one is only allowed to cut the rectangle into convex polygons.
8. Introduce and estimate a similar function, when one is allowed to cut the rectangle into any figures whose interiors are open and connected.
9. Suggest and study additional directions of research.

4. Guessing Bracket Sequences

A *balanced bracket sequence* is a string consisting of only brackets, such that this sequence, when inserted certain numbers and mathematical operations, gives a valid mathematical expression. For instance $(())()$ is a balanced bracket sequence, but $(())($ is not.

Maksim is playing the following game. Let k be a positive integer. The goal is to guess a balanced bracket sequence, if it is known that

- a) the length of the sequence is $2k$;
- b) the length of the sequence is at most $2k$ (note that if Maksim asks the host to reveal a bracket at a position where there isn't any, the host replies with "no bracket").

Each move Maksim can ask the host to reveal the bracket at any position. Maksim wins when he correctly guesses the bracket sequence.

1. Let $M(k)$ be the minimal number of moves with which Maksim can guarantee a win regardless of the initial sequence. Find $M(k)$ or estimate it.
2. Now, assume that with every bracket the host also reveals the bracket that comes in pair with it. Let $L(k)$ be the minimal number of moves with which Maksim can guarantee a win regardless of the initial sequence. Find $L(k)$ or estimate it.
3. If the bracket sequence is not necessarily balanced, how many moves, in terms of k , does Maksim need

- a) to check whether or not it is balanced?
- b) to correctly guess the bracket sequence?

4. Now, Maksim can also ask whether a certain subsequence of the bracket sequence forms a balanced bracket sequence, that is, he can ask whether the substring from a -th to b -th bracket ($a \leq b$) forms a balanced bracket sequence. How many moves, in terms of k , does Maksim need to correctly guess the bracket sequence?

5. Suggest and study additional directions of research.

5. Random Graphs

Let $G = (V, E)$ be a finite undirected connected graph. Let $p \in [0, 1]$ and $\alpha, \gamma \in]0, 1[$. We consider a random graph $G' = (V, E')$ obtained from G by deleting each edge with probability $1 - p$. Let $W(G, p)$ be the size of the largest connected component in G' . Define the critical probability $p_c(G, \alpha, \gamma)$ as follows

$$p_c(G, \alpha, \gamma) = \inf\{p \in [0, 1] : \mathbb{P}(W(G, p) \geq \gamma \cdot |V|) \geq \alpha\}.$$

1. Let G be the 3×3 -grid (that contains 16 vertices). Find $p_c(G, \alpha, \gamma)$.
2. Find $p_c(G, \alpha, \gamma)$ when G is the complete graph on n vertices.
3. Let G_n be the $n \times n$ grid. Is that true that $\lim_{n \rightarrow \infty} p_c(G_n, \alpha, \gamma) = \frac{1}{2}$ for any fixed α, γ ?
4. Let $d > 2$ be a positive integer number and G_n be the d -dimensional $n \times \dots \times n$ grid. Estimate $p_c(G_n, \alpha, \gamma)$.
5. Let $G_n = A_n$ be the graph of the permutohedron of order n . The *permutohedron* A_n is a polytope in $\mathbb{R}^n$, the vertices of which are formed by permuting the coordinates of the vector $(1, 2, \dots, n)$, and two vertices $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are connected if and only if one is obtained from the other by interchanging the numbers i and $i + 1$ for some $1 \leq i \leq n - 1$. Is it true that $\lim_{n \rightarrow \infty} np_c(G_n, \alpha, \gamma) = 1$ for any α, γ ?
6. Determine whether it is possible to construct a graph G such that $p_c(G, \alpha, \gamma) = p$ for given p, α, γ .
7. Suggest and investigate your directions of this problem.

6. Functions on Positive Integers

We call a function $f : \mathbb{N} \rightarrow \mathbb{C}$ *nice* if $f(mn) = f(m) + f(n)$ for all positive integers m and n .

1. Find all nice functions such that $|f(n)|$ is monotonically decreasing.
2. Find all nice functions such that $|f(n)|$ is monotonically increasing.
3. Let $a_1, \dots, a_k$ be fixed positive integers, and let f be a nice function. Define $g(n) = \max\{|f(n + a_1)|, \dots, |f(n + a_k)|\}$ for every $n \in \mathbb{N}$. Find all nice functions f , for which the function g is monotonically decreasing.

We call a function $f : \mathbb{N} \rightarrow \mathbb{R}$ *interesting* if $f(mn) = f(m)f(n)$ for all *coprime* positive integers m and n .

4. Let $a_1, \dots, a_k$ be distinct positive integers. For an interesting function f , consider the function $g(n) = \max\{f(n), f(n + a_1), \dots, f(n + a_k)\}$ for every $n \in \mathbb{N}$. If g is a strictly monotonic function, then prove that $f(n) = n^C$ for some real number C .

5. What if instead of being strictly monotonic on the entire $\mathbb{N}$, the function g is monotonically increasing on a subset of $\mathbb{N}$ of density 1?
6. Suggest and study additional directions of research.

7. Compositions of Polynomials

For a polynomial $P(x)$ with coefficients in a ring R write $P^{m+1}(x) = P(P^m(x))$ for any positive integer m . Recall that a polynomial is *monic* if its leading coefficient is equal to 1. Throughout this problem all polynomials are assumed to be non-constant.

1. Suppose $P(x)$ has rational coefficients and degree at least two and $P(P(x))$ has integer coefficients.
 - a) Is it true that $P(x)$ also has integer coefficients?
 - b) What if we impose the additional condition that $P(0)$ is an integer?
2.
 - a) Determine all pairs (a, b) of positive integers with the following property: any polynomial $P(x)$ with rational coefficients, such that $P^a(x)$ and $P^b(x)$ have integer coefficients, also has integer coefficients.
 - b) What happens if we require that the statement holds only for all polynomials of degree at least 1000?
3. Suppose $P(x)$ and $Q(x)$ are two polynomials of different degrees and have rational coefficients but $P(Q(x))$ has integer coefficients. Is it always true that at least one of $P(x)$ and $Q(x)$ has integer coefficients?
4. Suppose $P(x)$ and $Q(x)$ are two polynomials with the same degree and have rational coefficients but $P(Q(x))$ and $Q(P(x))$ have integer coefficients. Is it always true that at least one of $P(x)$ and $Q(x)$ has integer coefficients?
5. We call a monic polynomial $P(x)$, of degree at least 4, *monic decomposable over R* if $P(x) = A(B(x))$ for some monic polynomials $A(x), B(x) \in R[x]$ of degree at least 2.
 - a) Prove that if $P(x) \in \mathbb{Z}[x]$ is monic decomposable over $\mathbb{C}$, then it is also monic decomposable over $\mathbb{Z}$.
 - b) Give a polynomial $P(x)$ with coefficients over $\mathbb{F}_2$ which is monic decomposable over a field containing $\mathbb{F}_2$, but is not monic decomposable over $\mathbb{F}_2$.
6. Suppose $Q(x)$ has integer coefficients, $Q(0) = 0$ and the greatest common divisor of the coefficients of $Q(x)$ is 1. Suppose that $P(x)$ has complex coefficients and $P(Q(x))$ has integer coefficients. Prove that $P(x)$ also has integer coefficients.
7. Formulate and investigate similar questions for a polynomial $P(x, y)$ composed with polynomials $f(x)$ and $g(y)$, that is, given an assumption on the coefficients of $P(x, y)$, $f(x)$, $g(y)$ and $P(f(x), g(y))$, investigate whether a stronger conclusion for the coefficients must hold.
8. Suggest and study additional directions of research.

8. Midpoints and Angles

Let k be a positive integer, and let $\alpha \in \left]0, \frac{\pi}{2}\right]$ be a positive real number. Let S be a non-empty set of distinct points in the plane. For any two distinct points A and B from S , call a line *good for AB* if it passes through the midpoint of the segment AB and determines angle α with AB . There are two such lines when $\alpha \neq \frac{\pi}{2}$ and one such line when $\alpha = \frac{\pi}{2}$. Call a point from S *special for AB* if it lies on a line which is good for AB . We are interested in sets S with

the following property: for any two distinct points A and B there are exactly k points from S which are special for AB . Such sets will be called (k, α) -symmetric.

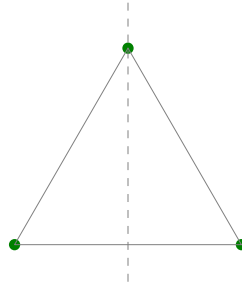


FIGURE 3. Vertices of an equilateral triangle form a $(1, \frac{\pi}{2})$ -symmetric set.

1. Suppose $k = 1$ and $\alpha = \frac{\pi}{2}$.
 - a) Can S be infinite?
 - b) Can S be finite, but arbitrarily large, that is, for any positive integer N , there exists a finite set S with at least N points? If yes, prove it. Otherwise, give an upper bound on the number of points in S .
 - c) Find all integers $n \geq 3$ for which there is such a set S with n points.
2. Answer the questions in Part 1 for
 - a) $k = 1$ and any α ;
 - b) $k = 2$ and $\alpha = \frac{\pi}{2}$.
3. Answer the questions in Part 1 for as many other k and α as you can.
4. Generalize the problem to three dimensions, by considering sets S with not all points lying on a plane. Feel free to substitute some of the wording in the original statement, by appropriately working with angles between two lines, or between a plane and a line, or between two planes.
5. Suggest and study additional directions of research.

9. Sequences Generated by Constants

Let $(a_n)_{n \geq 1}$ be a strictly increasing sequence of positive integers.

1. In this part suppose $a_{n+1} \leq 2a_n - 2$ for all $n \geq 1$.
 - a) Prove that for any such sequence there is a constant b_1 such that the sequence $(b_n)_{n \geq 1}$ of positive real numbers with $b_{n+1} = [b_n](b_n - [b_n] + 1)$ satisfies $[b_n] = a_n$ for all $n \geq 1$. Give an explicit expression for b_1 depending only on $(a_n)_{n \geq 1}$.
 - b) With the additional assumption $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$, prove that b_1 is irrational. Does this hold if we drop the assumption?
 - c) With the additional assumption $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$, is it always true that there does *not* exist a polynomial $P(x)$ with integer coefficients such that $P(b_1) = 0$? If yes, prove it. Otherwise, give an example for $(a_n)_{n \geq 1}$ when $P(x)$ exists and an example when it does not exist.
2. Does part 1a) always hold if the inequality $a_{n+1} \leq 2a_n - 2$ is replaced by $a_{n+1} \leq Ka_n$, where:
 - a) $K = 2$;
 - b) $K = 2 + \frac{1}{1000}$?

3. a) Suppose that $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$ for all $n \geq 1$. Is it true that there exists a constant C such that $a_n = \left\lfloor C^{2^n} + \frac{1}{2} \right\rfloor$ for all $n \geq 1$?
- b) Suppose that $a_{n+1} \geq a_n^2 - a_n + 1$ for all $n \geq 1$ and that $\lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{a_n}$ is a rational number. Is it true that $a_{n+1} = a_n^2 - a_n + 1$ for some N and all $n \geq N$?
- c) Suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n^2} = 1$ and that $\lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{a_n}$ is a rational number. Is it true that $a_{n+1} = a_n^2 - a_n + 1$ for some N and all $n \geq N$?
4. Suggest and study additional directions of research.

10. Averaging Functions

Let E be the set of all real bounded sequences. For $x, y \in E$ put $d(x, y) := \sup_{n \in \mathbb{N}} |x_n - y_n|$. We say that a sequence of sequences $\{x^{(k)}\} \subset E$ *approximates* a sequence $y \in E$, if the sequence of real numbers $d(x^{(k)}, y)$ tends to 0 as $k \rightarrow \infty$.

We call a sequence of non-empty finite subsets $A_n \subset \mathbb{N}$ *absorbing* if

$$(1) A_n \subseteq A_{n+1} \quad \text{and} \quad (2) \bigcup A_n = \mathbb{N}.$$

For any such sequence $A = \{A_n\}$, we define the *averaging function* $C_A : E \rightarrow E$ associated to it as follows: for $x \in E$ its image $C_A x$ is the sequence with

$$(C_A x)_n := \frac{1}{|A_n|} \sum_{i \in A_n} x_i \quad \text{for every } n \in \mathbb{N}.$$

Let us fix an absorbing sequence $A = \{A_n\}$ and its averaging function C_A .

1. Check that d satisfies the conditions of a metric function (non-negativity, symmetry, the triangle inequality and that $d(x, y) = 0$ if and only if $x = y$).
2. Check that C_A is well-defined (that is, if $x \in E$, then $C_A x$ is also in E).
3. Check that if $x \in E$ has a limit, then $C_A x$ also has the same limit.
4. Check that if a sequence of sequences $\{x^{(k)}\} \subset E$ approximates $y \in E$, then $\{C_A x^{(k)}\} \subset E$ approximates $C_A y \in E$.
5. Under which conditions imposed on A , can we conclude that the function C_A is a) injective? b) surjective?
6. Let $\{A_n\}$ and $\{B_n\}$ be absorbing sequences, and let C_A and C_B be the corresponding averaging functions. Try to find under which conditions imposed on $\{A_n\}$ and $\{B_n\}$, for any $x \in E$, one has $\lim_{n \rightarrow \infty} ((C_A x)_n - (C_B x)_n) = 0$.

For a subset $A \subset \mathbb{N}$, one can define its *boundary*

$$\text{bd}(A) := \{n \in \mathbb{N} \mid n \notin A \text{ and } (n-1 \in A \text{ or } n+1 \in A)\}.$$

An absorbing sequence $\{A_n\}$ will be called *good* if $\lim_{n \rightarrow \infty} \frac{|\text{bd}(A_n)|}{|A_n|} = 0$.

7. Let $x \in E$ be such a sequence that, for any absorbing sequence $A = \{A_n\}$, the averaged sequence $C_A x$ converges. What can you say about x ?
8. Let $x \in E$ be such a sequence that, for any good absorbing sequence $A = \{A_n\}$, the averaged sequence $C_A x$ converges. What can you say about x ?
9. Let $x \in E$. Consider the following sequence of sequences $\{x^{(k)}\} \subset E$: $x^{(1)} = x$ and $x^{(k+1)} = C_A x^{(k)}$ for $k \geq 1$. For which $x \in E$ this sequence approximates a sequence?
10. Suggest and investigate your own directions of research. For example, describe situations in which the sequence $x \in E$ does not have a limit, but its averaged sequence $C_A x$ does.

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